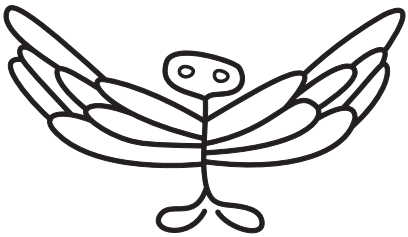
unwoven

Community-driven Furniture Production
Through Reframing Waste Materials

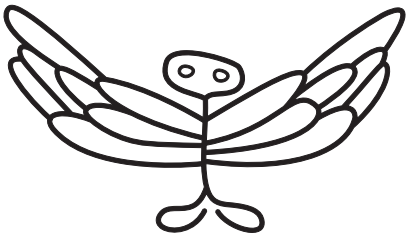


Designed by Wren Pearson



Chapter 1

To create is a
beautiful thing



Unwoven

Unwoven is a project that aims to return the power of manufacturing and craft to the end user of the product. It encourages you to take an active part in the making of a stool, reimagined through the lens of responsible material use within a circular economy.

You are provided with a set of instructions and patterns for building a mould that will allow you to create a stool from recycled textiles, with a particular focus on cotton and denim.

This stool is a purposefully transient object. It will wear and break down with time and exposure to the elements, but it's constituent materials remain strong, allowing it be destroyed and rebuilt again.

This project can be completed with mostly simple tools and store-bought materials, but a level of experience with power tools and DIY is important to guarantee success and safety.

From start to finish, the project is expected to take 2-4 weeks.



Tools

- Jigsaw
- Scroll cutting blade
- Long jigsaw blade (>100mm)
- Electric drill
- 3mm drill bit
- Caulking gun
- Heavy duty stapler
- Rotary cutter
- Fabric scissors
- Cutting mat
- Blender or grain mill
- Large bucket for mixing
- 3D Printer
- A4 Printer
- Safety glasses
- Dust mask

Don't use anything that will get used to make food again!


DEWALT
Makita

Materials

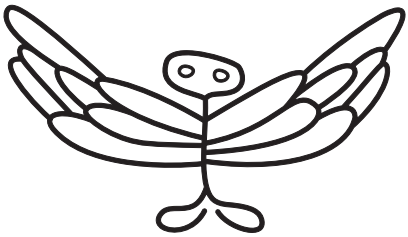
- 12mm plywood (min 1200x900)
- 50mm extruded polystyrene (min 1200x600)
- Spray adhesive
- Liquid nails
- Timber screws (20-40mm)
- Heavy plastic sheeting (1m^2)
- Packing tape
- Masking tape
- Sandpaper (80-120 grit)
- Cotton fabric
- Carboxymethyl Cellulose (CMC) (400-500g)
- String or rope (~1m)


Super
77
Spray
Adhesive



Chapter 2

Making things to make things



Print patterns

To begin making the moulds for the Unwoven Stool, download the tiled patterns from Github. Print all the pages for both patterns (ensuring your printer is set to 100% scale by checking the 100mm reference line on the first page).

Once all the pages are printed, lay them out on a flat surface and carefully line them up and tape them together to create a full sized pattern.

Begin 3D printing

At this point you should also begin printing the 4 parts of the mould that need to be 3D printed. These files are available on Github in a range of file formats to best suit your printer.

If you don't have a 3D printer, don't worry! Check if you have a local library or makerspace with 3D printers you can access. Otherwise search for a supplier either locally or internationally, you will likely find a way to print these parts for only a small fee.

Glue patterns down

Once you have 2 full sized patterns taped together, it is time to stick them down.

Make sure you stick the correct pattern to the correct material!

If you were unable to get either the foam or ply in the sizes listed on page 5, you can split the patterns across multiple pieces of material as needed.

I prefer to use a spray adhesive to stick the patterns down, but a glue stick or double-sided tape will also work.

Spray the adhesive liberally on both the material and the pattern, then leave them seperate for a few minutes until they are tacky to the touch. Bring the two together and press down firmly to ensure a strong bond. Leave to cure for 30-60 minutes before you begin cutting.

Cut plywood

It is now time to cut out the plywood parts using the pattern we stuck down.

A jigsaw is needed for this, and a blade specifically designed for scroll cuts (tight curves) can be helpful if you have one.

You will also need a drill with a 3mm drill bit to make the pilot holes that will help with assembling the moulds later. I suggest drilling all the holes as marked before cutting the pieces out. This way the locations won't be lost if any of the pattern is damaged during the cutting process.

It can be helpful to roughly cut out the shapes from the sheet of plywood first, and then trim them to their final size separately.

After cutting out each piece, I recommend peeling away the paper pattern and labelling it with a pencil. This will help to keep all of the pieces neatly organised.

Cut foam

Cutting the foam pieces is essentially the same process, but you will need to make sure to use a longer blade for the jigsaw (over 100mm) to cut through the thicker foam, especially for the angled cuts.

Always wear a dust mask when cutting or sanding the foam!

Cut all the pieces out roughly from the board first, and then trim them to size along the square edges, leaving the angled cuts for now.

A simple protractor can help with setting the angle of your blade if your jigsaw doesn't have small enough increments on the angle adjustment dial.

Cut all of the angles in groups to get a more consistent result. For example, set your jigsaw blade to 5 degrees and cut all of the 5 degree bevels at once. This way, even if your blade is not set to exactly 5 degrees, the inaccuracy will be consistent across all the parts.

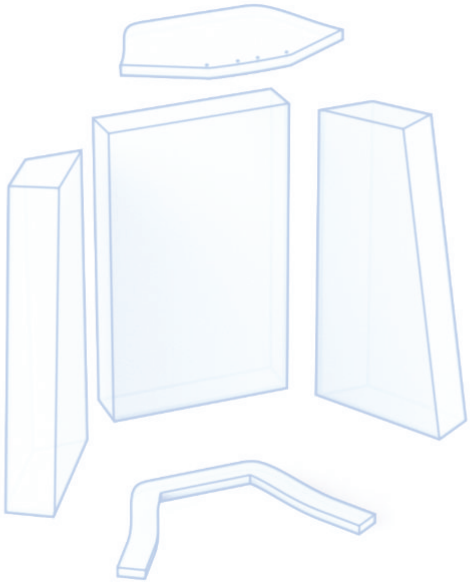
Once again, label all the parts with a pencil as you cut them.

Fit parts together

It is time to begin attaching the parts together. Begin by laying out all the pieces for the inner and outer moulds respectively, setting aside the plywood pieces labelled Bottom Overall, Front Overall and Side overall, including their accompanying insert pieces.

Take the Outer Bottom and lay it flat on a table. Position the 3 foam pieces on top of it as shown in the illustration on the next page, making sure that the face you laid the pattern on is facing to the outside of the curve. Then place the Outer Top on top of the foam, making sure it is centred to the Outer Bottom. When you are happy that everything fits, take the pieces apart and put liquid nails on the surfaces to be glued together. Carefully reassemble all the pieces and use masking tape to hold them in place while the glue sets.

Repeat this process for the inner mould and then leave the glue to set overnight.



Sand foam down

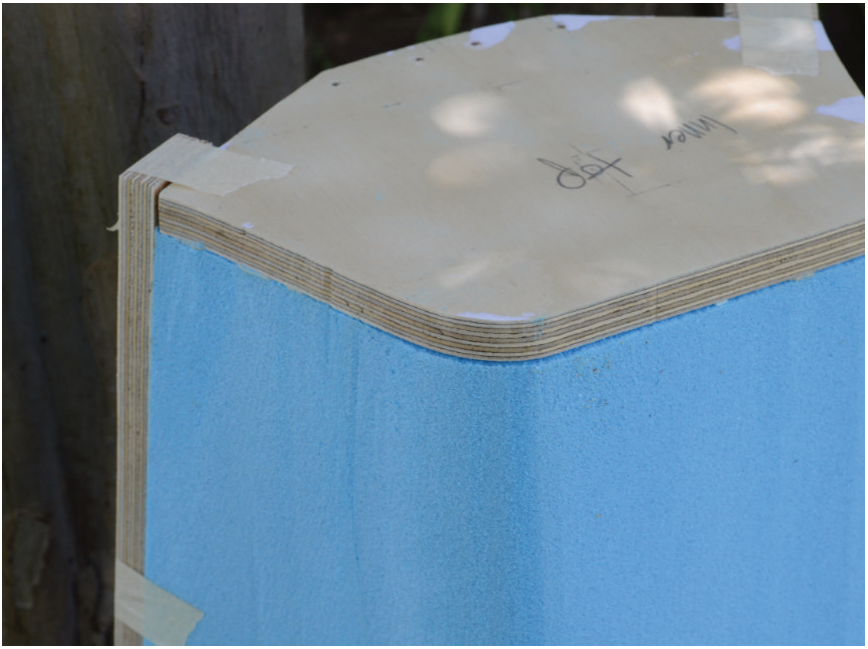
Once the glue has dried, the foam pieces need to be sanded back to the profile of the plywood pieces they are attached to.

To make this easier, I suggest making a sanding stick out of some scrap plywood. Cut a rectangular piece of plywood about 100mm wide and 500-600mm long. Then stick down a strip of 80 or 120 grit sandpaper to it with double-sided tape or spray adhesive.

You may also need to make a second sanding stick to get into the internal curves of the mould. A piece of PVC pipe or a wooden dowel is perfect for this.

Use these sanding sticks to gently remove material from the foam. Do not push too hard or fast as you will start to tear the foam and create an uneven surface. The length of the sanding sticks you made will let you use the plywood pieces as a reference point for when the surface is perfectly flat.





Attach front plywood pieces

The inner and outer moulds have two front pieces each that were cut from plywood earlier.

When the foam has been sanded down, use liquid nails to attach these pieces securely. Hold them in position with masking tape while they set.



Attach plywood inserts

The left and right Front Overall pieces, as well as the Bottom Overall piece, have inserts that need to be glued and screwed together.

It is important that the inserts get attached to the back face of these parts (not the side that the pattern was laid on).

Line up the pilot holes on both parts to ensure they are positioned correctly.



Wrap moulds with plastic

It is now time to wrap the inner and outer mould sections with heavy duty plastic sheeting (sometimes called builders plastic). This will help to waterproof the mould and prevent the cotton material from sticking as it dries.

The plastic also serves as a sacrificial layer that can be replaced if it gets damaged, prolonging the life of the mould.

To attach the plastic you will need either a heavy duty staple gun or upholstery tacks.

Start by attaching the plastic along the left edge of the mould and pull the plastic tight as you move right across the mould, tacking it down as you go.

Don't worry if it's not perfect, the mould will still work perfectly fine with some wrinkles or imperfections in the plastic.





Attach 3D prints and assemble

The moulds are almost finished. You can now attach the 3D printed components to the rest of the mould.

The Top 3D prints attach to the piece of ply labelled Top Overall. The pilot holes in the ply should line up with the holes in the prints. Use liquid nails and screws to secure them in place.

The Front 3D prints attach to the top of the inner mould. They are secured in the same way.

To prevent the cotton material from sticking, wrap the 3D prints and any exposed ply with a layer of packing tape. This will also disguise the seams between the halves of the prints.

Finally, you can assemble all the pieces of the mould with screws. Avoid using any glue for this as you will need to take the mould apart to get the stool out later. Leave the top of the mould seperate for now so that you can fill it with cotton easily.

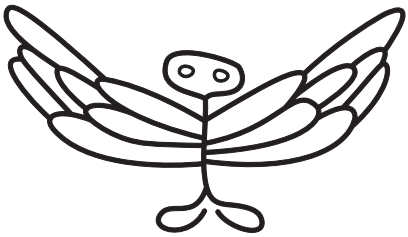






Chapter 3

A slimy mess

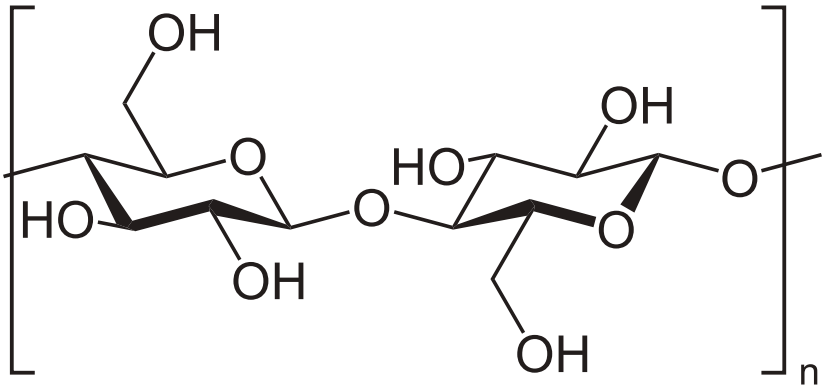


Carboxymethyl Cellulose

Carboxymethyl Cellulose (CMC) is a water-soluble cellulose polymer derived from natural sources such as wood pulp and cotton fibres. It is commonly used as a thickener and stabiliser in the food and cosmetics industries.

The Unwoven stool uses a 4% CMC solution as a binder to transform shredded cotton fibres into a solid material.

Although CMC is generally food-safe and quite harmless, it can be a slimy and uncomfortable material to work with, so I would suggest wearing gloves whenever dealing with it.



Mixing CMC

An entire Unwoven stool will require about 12L of 4% CMC solution, 6L per side.

To make this more manageable, I mix CMC 1.5L at a time. Simply measure out 1.5L of water and 60g of CMC powder and mix them together.

The solution will quickly become thick and slimy, and some lumps will form that are difficult to mix. Just leave the solution to dissolve fully for a day or two and you should be left with a thick, slightly yellow gel.



Preparing the fabric

Before it is combined with the CMC solution, the fabric must be cut down and shredded. To make one unwoven stool, you will need 8-9 pairs of jeans worth of cotton.

I recommend cutting the fabric down with a rotary cutter and a cutting mat, making sure to remove any zippers or buttons. Cut the fabric into small squares 2-3cm wide.

To fully shred the fabric you will need access to either a small grain mill or a powerful blender. Neither of these should be used to process food again after shredding fabric.

A small grain mill is the most effective method and they can usually be bought for less than $70 AUD online. They often lack any safety features though and should be used with appropriate caution.

If you only have access to a blender, you will need to shred the fabric in a slurry of water to prevent overloading the motor. Make sure to dry the shredded fabric completely before mixing it with the CMC solution.

Mixing and moulding

To create the CMC-cotton mixture that will form the stool, begin mixing a few large handfuls of shredded cotton with about a litre of CMC in a large bucket.

The exact ratio is not vital, but the fully combined mixture should have CMC mixed evenly throughout, with no dry cotton and not too much excess CMC that isn't connected to cotton fibres.

Add more CMC or cotton as required and mix thoroughly, as the cotton will likely form clumps that will appear to be coated but are actually dry inside when you pull them apart.

Once you have a bucket mostly full of CMC-cotton mixture, begin filling the mould, packing the mixture in tightly.

Repeat this process until the mould is completely full, then press the top of the mould down firmly and clamp or screw it into place.

Drying process

With the mould fully closed, the material will dry very slowly, but I still find it best to keep it closed for 12-24 hours before opening it again, as this helps it hold it's form better as it dries.

After the first 12-24 hours, open the top of the mould and leave it outside in direct sunlight. If you can't leave it in the sun, you can also place it in front of a small heater or even a fan.

After another few hours, unscrew the front panels and remove them. If any cotton fibres stick to the mould and damage the surface of the stool, just gently smooth out the area with your finger and add more cotton mixture if needed.

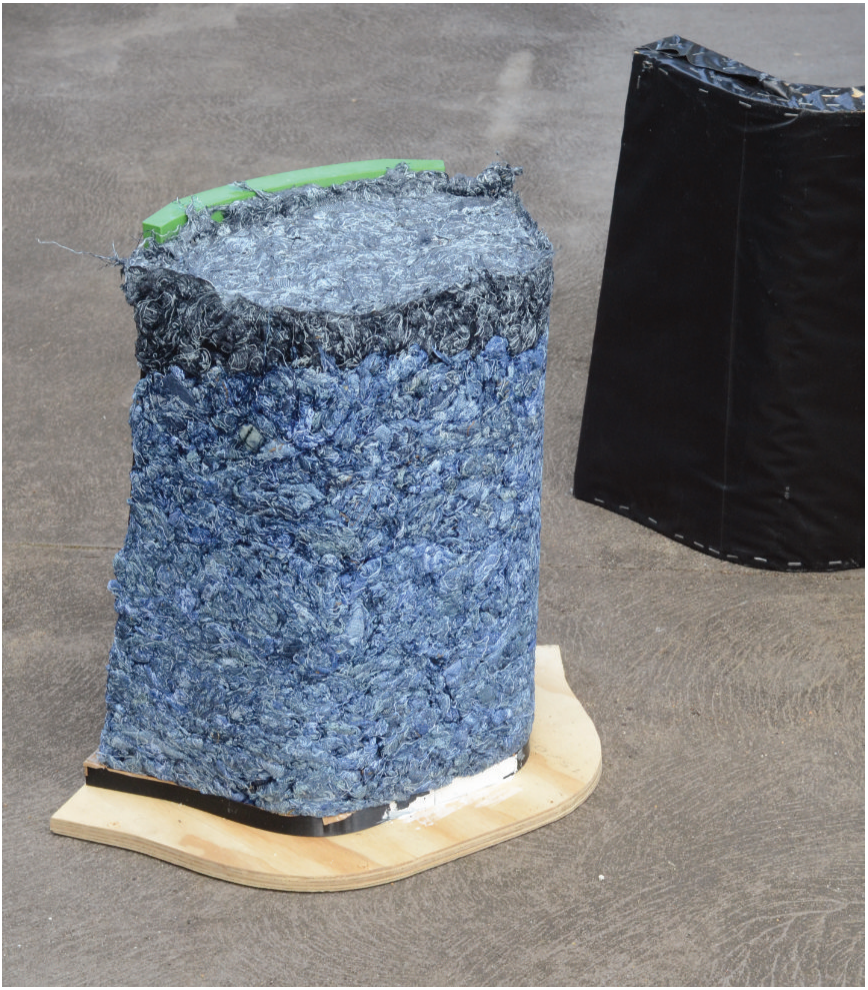
When the exposed surfaces are dry to the touch, unscrew the outer mould and gently remove it, prying it apart from the stool slowly to prevent any damage.

After 1-2 more days of drying, you can transfer the stool back into the outer mould while removing it from the inner mould.

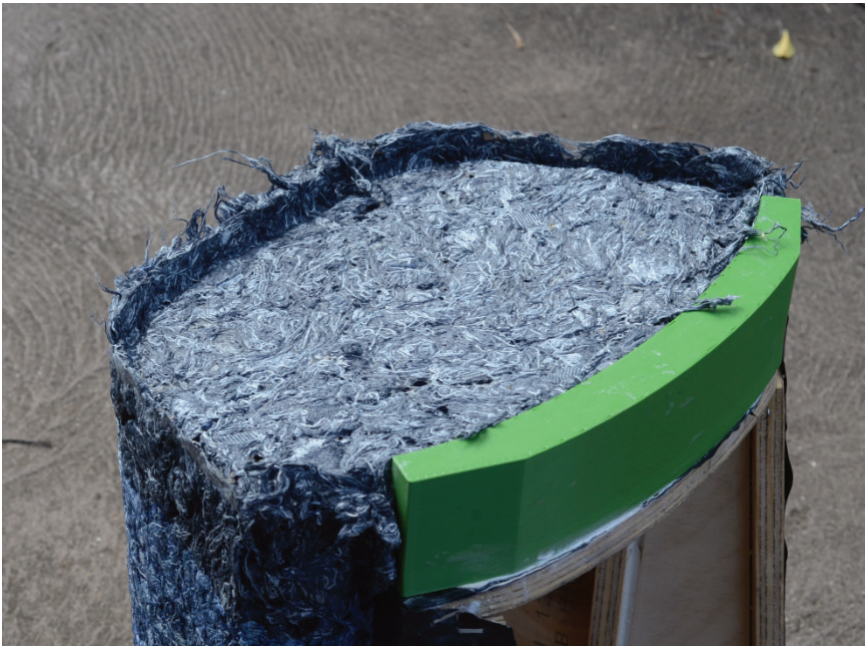
Leave it like this for 1-2 days and then remove it from the mould completely, allowing it to dry fully while standing up.

Repeat

You now have 1 half of an Unwoven stool. Reassemble the mould and repeat the process of filling it and drying it.








click
CERAMIC HEATER


click



Combine the halves

Once you have two completely dry halves, you can trim any excess fabric and attach them together to create the final stool.

Use a sharp knife to trim off any excess fabric leftover from the moulding process. Then simply drill or poke 8 holes through each half where they meet underneath the seat, and thread a string, rope or ribbon through them, tying it tightly to keep them securely together.

Congratulations!

You have made an Unwoven stool!

Remember to keep your stool in a dry environment, as the CMC remains water-soluble. If your stool is ever damaged or is no longer needed, you can soak it in water and the CMC will wash away, leaving only the shredded cotton, which can be recycled or moulded into a new stool.





